

ORIGINAL ARTICLE

How Accurate Is Inpatient Blood Pressure Measurement?

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BACKGROUND: Accuracy of blood pressure (BP) measurement depends on quality technique, yet there are no existing guidelines for accurate measurement of inpatient BP. We assessed the performance of BP measurements in a major teaching hospital by comparison with consensus criteria.

METHODS: Routine BP measurement methods were observed across 3 noncritical care wards (aged care, acute surgical, and hematology) at St George Hospital, Sydney, over a 3-week period. Aspects of BP measurement were compared for concordance with an ideal BP measurement technique, consisting of 10 consensus criteria, as determined by an expert panel.

RESULTS: A total of 278 BP measurements were observed in 153 patients. Concordance with the consensus criteria was high for device calibration (92%), uncrossed legs (82%), using the correct cuff size (77%), and back support (75%). Concordance was moderate for the criteria of correctly positioned cuff on a bare arm (both 65%), and poor for the remaining criteria (seated position [26%], no talking [23%], cuff at heart height [15%], and taking at least 2 measurements [7%]). No observed measurement met all 10 of the consensus criteria for BP measurement quality.

CONCLUSIONS: Our findings indicate considerable discordance between routine BP measurement and optimal technique (as determined by an expert panel) in an Australian hospital setting. The lowest concordance was observed for the 3 highest-ranked consensus criteria for accurate BP measurement. This variability may result in inaccurate and biased readings. Standardized, hospital-specific BP measurement guidelines and training are needed to improve BP measurement accuracy. (*Hypertension*. 2026;83:00–00. DOI: 10.1161/HYPERTENSIONAHA.125.26355.) • [Supplement Material](#).

Key Words: aged ■ blood pressure ■ hematology ■ humans ■ leg

Accurate blood pressure (BP) measurement depends on multiple factors, including using an appropriate validated device, observers undertaking correct methods, and proper patient preparation and positioning. Failure to ensure a standardized and quality technique risks an inaccurate BP readings, with currently unquantified implications for patient care in the hospital setting, but with foreseeable risks of erroneous treatment decisions.

Current guidelines advise outpatient assessment of BP under ideal or standardized conditions,^{1–4} but there is little empirical evidence on how to correctly measure BP

in the inpatient setting.^{5–7} Low BP in an admitted patient is a useful sign of acute clinical deterioration,^{8,9} but the significance of elevated asymptomatic BP is less clear in both acute and longer-term scenarios.⁷ To our knowledge, there is no evidence on optimal measurement techniques in hospital settings.¹⁰ As a result, there are currently no national or international standards to guide routine inpatient BP measurement.

In the context of uncertain standards against which inpatient BP technique should be measured, we conducted a literature review and Delphi survey to identify the key variables in the accurate measurement of

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NOVELTY AND RELEVANCE

What Is New?

To our knowledge, this is the largest published audit characterizing blood pressure (BP) measurement techniques during routine inpatient care that is rated against an expert panel-derived criteria of what constitutes a quality BP measurement technique.

What Is Relevant?

This is the first study to systematically examine how BP is actually measured during routine hospital care and identify which technical errors are most common.

Clinical/Pathophysiological Implications?

When BP is not measured properly in hospitals, treatment decisions can be based on false information. This can cause harm and is why this study is a step toward bridging the gap and working toward better quality standards in BP measurement.

Nonstandard Abbreviations and Acronyms

BP	blood pressure
SBP	systolic blood pressure
WA	Welch Allyn

inpatient BP and their ranked importance, which allowed us to define a set of expert consensus criteria against which to assess the performance of BP measurement in hospital inpatients (Figure 1).¹¹

The present study was designed to audit real-world inpatient BP measurement techniques at a large tertiary referral hospital in Sydney, Australia, and to assess these techniques against consensus criteria agreed upon by an expert panel.

METHODS

Data Availability

Anonymized data and materials have been made publicly available at Zenodo and can be accessed at <https://doi.org/10.5281/zenodo.17509956>.

Consensus Criteria for Optimal BP Measurement Technique

A Delphi survey of an expert panel of 9 BP specialists (comprising 3 dedicated researchers and 6 clinician-researchers, all nephrologists) was used to derive the ranked list of 10 key criteria recommended for accurate BP measurement in admitted patients.¹¹ The panel collectively had >2300 publications in cardiovascular/BP research, and all had contributed to hypertension guideline development. The consensus process included ranking each criterion in descending order of expected impact on accuracy (Figure 1). This formed the consensus criteria against which observed BP measurement practices were measured as defined in Table 1. These consensus criteria in Table 1 were determined through a 2-stage Delphi process in accordance with ACCOR and CREDES

guidelines.^{14,15} Stage 1 involved document ratification (>80% agreement threshold) to validate source documents. Stage 2 involved 3 iterative rounds of variable ranking: independent ranking of 10 variables (Round 1), re-ranking following anonymized feedback (Round 2), and paired comparisons to refine rankings (Round 3). Consensus was defined a priori as a Kendall W coefficient ≥ 0.7 . Full Delphi panel composition is provided in Table S1.

Audit of Routine Inpatient BP Measurement techniques

The audit was conducted between September 15, 2023 and February 22, 2024 on 3 wards (acute aged care, hematology/oncology, and acute surgery) at St George Hospital, a 627-bed tertiary referral hospital in metropolitan Sydney, Australia. The acute aged care ward is an inpatient ward for geriatric patients with acute medical illnesses. A single independent auditor (AG) attended each ward over 12 sessions (6 morning and 6 afternoon) to observe routine collection of vital signs. Observation days were standardized to the same days of the week for each ward to account for day-based workload variation. All routine assessments of BP were eligible for observation and inclusion, regardless of the clinical position of the staff member. No BP measurements were excluded from the audit based on clinical contraindications to consensus-concordant technique. This was a deliberate methodological choice to reflect real-world hospital practices and examine the feasibility of adopting consensus criteria as a practice standard. Potential clinical impediments to optimal technique, including patient mobility limitations, anatomic contraindications, and behavioral factors were recorded. In addition to recording the occurrence (or not) of the 10 reference consensus criteria, additional descriptive patient data were obtained in a de-identified manner (Figure S1). Verbal consent to observe clinical procedures was obtained from each staff member and patient at the time of data collection. To minimize changes in usual practice, neither the staff member nor the patient was made aware that the focus of the audit was BP measurement. Correct cuff sizing was verified by the auditor as detailed in Table 1. The audit methodology, including incomplete disclosure, to participants, was approved by the local ethics committee (South Eastern Sydney Local Health District 2020/STE02888).

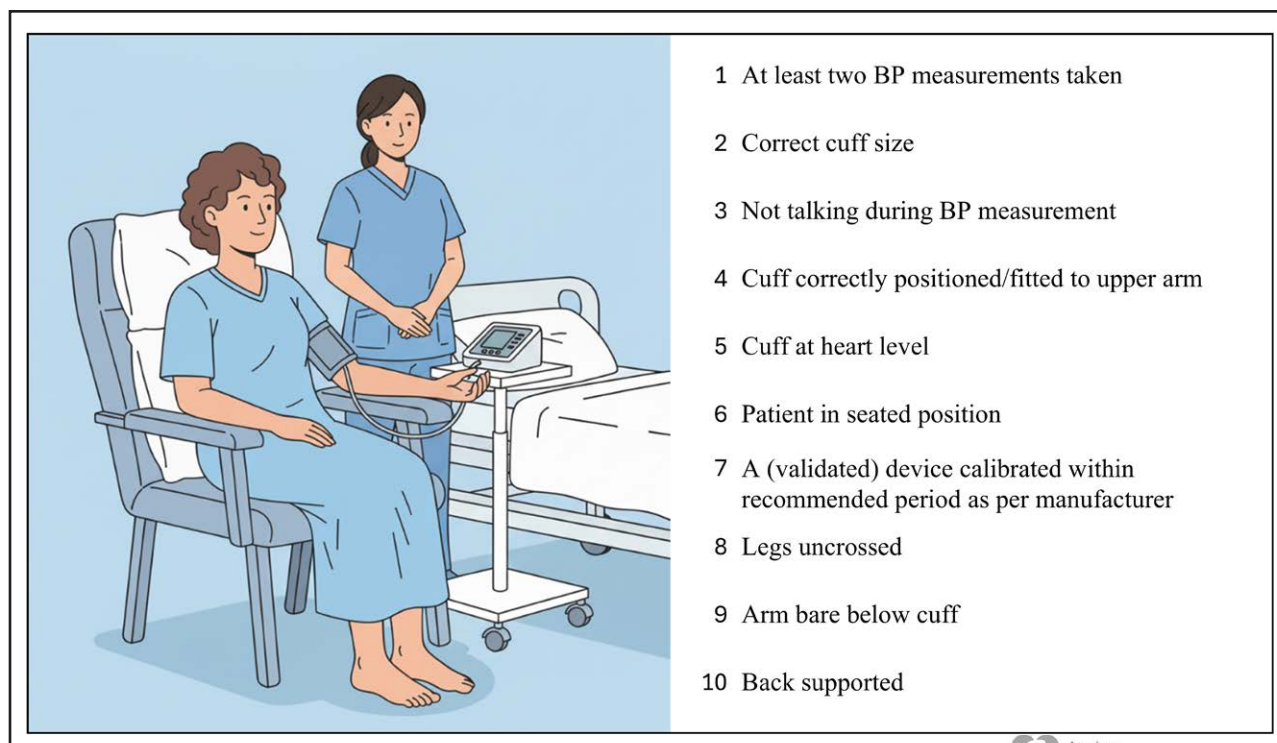


Figure 1. Consensus criteria for accurate inpatient blood pressure (BP) measurement as per expert panel, listed from most important to least important (with accompanying visual representation).

Image generated using Gemini AI software.

Statistical Analysis

Continuous variables were described using mean and SD or median and interquartile range for skewed data. Categorical variables were presented as number and proportion of the total. The proportion of BP measurements achieving each criterion was calculated. Concordance with consensus criteria was defined as good ($>70\%$), moderate ($30\%–70\%$), or poor ($<30\%$). These thresholds were subjectively defined a priori after consideration of relevant quality improvement and audit manuals.^{16–18} Criteria had been ranked previously by the expert panel from most important to least important for accuracy.

As no BP measurements met all 10 criteria, multivariable logistic regression was replaced with subgroup analysis examining associations between patient, environmental, and observer-related independent variables, with formal testing of coefficient equality for statistical significance. Arm circumferences were grouped into 3 categories, aligning with standard BP cuff sizes. Systolic BPs (SBPs) were grouped to match locally mandated clinical deterioration parameters, with a >160 mm Hg category to capture grade 3 chronic hypertension criteria based on 24-hour ambulatory readings as a more sensitive threshold (SBP <100 mm Hg, SBP 100 to 159 mm Hg and SBP >160 mm Hg).¹⁹ For context, an SBP below 100 mm Hg or >180 mm Hg mandates an escalation in clinical response in the observed hospital.^{9,20} Any staff member was eligible for inclusion in the audit. As only nurses, their assistants, and their students were captured, staff qualifications were graded by seniority: students, vocationally trained assistants and nurses, and senior staff (the latter comprising of registered nurses and clinical nurse consultants, both of whom have undertaken

university degree-based training, with additional postvocational training for consultant nurses).

We estimated a saturated linear probability model with criterion-specific indicators and no intercept (equivalent to a cell-means parameterization) so that each coefficient directly represents the observed compliance rate for that criterion. Patient-level cluster-robust standard errors were used to account for repeated observations, ensuring valid inference under within-patient correlation. To assess whether compliance patterns differed across clinical subgroups, the models were re-estimated separately by arm-circumference tertile, ward, SBP category, and observer experience level, and combined using seemingly unrelated estimation to test equality of coefficients across groups. Two-sided Wald tests evaluated the null hypothesis that coefficients were equal across all subgroup models—that is, that compliance did not differ between subpopulations. The resulting P values, shown in the figures, indicate the strength of evidence against this null: smaller values signify stronger support that at least 1 subgroup's mean adherence rate differs. $P < 0.05$ was considered significant.

RESULTS

Descriptive Analyses

In total, 278 BP measurements on 153 admitted patients were observed. Ten BPs were measured using auscultatory techniques, and the remainder (96%) were automated. In keeping with standard ward layout, most BP measurements were conducted in 4-bed rooms, although curtains formed a partial partition. A summary of the

Table 1. List of Consensus Criteria and Their Definitions

	Consensus criteria	Definition
1	At least 2 BP measurements taken	Observer measures BP at least twice, spaced 1 minute apart for the 1 occasion of routine vital sign observations
2	Correct cuff size	Use of appropriate cuff size by observer based on measurement of the largest diameter/midpoint of the acromion to olecranon of the upper arm to the nearest 0.5 cm by the independent auditor. NB: measurement completed after the observer completed BP measurement to avoid influencing technique
3	Not talking during BP measurement	Neither observer nor patient talking during BP measurement
4	Cuff correctly positioned/fitted to upper arm	Artery marker over the brachial artery, cuff snug with room for no >2 fingers underneath. BP cuff bladder width 37% to 50% of the patient's arm circumference (length to width ratio of 2:1)
5	Cuff at heart level	Cuff at the level of the right atrium/midpoint of the sternum as estimated by the independent auditor
6	Patient in seated position	Patient seated in a chair or upright on the bed with feet on the floor
7	A (validated) device calibrated within recommended period as per manufacturer	Electronic (oscillometric) device documented to have passed calibration† by local biomedical engineering team (Figure S2) in the last 12 mo as well as confirmed correct function at time of use including full functionality of keypad (if an automated device), intact hosing, cuff, connectors, plugs and sockets with no signs of damage. Device as a unit (monitor and paired cuff) has documented validation, as per international guidelines, publicly available either by manufacturer or independent authority* NB: for manual devices the same visual inspection and review of validation data applied, but no calibration records were available as these were not routinely conducted on these devices
8	Legs uncrossed	Legs do not crossover each other
9	Arm bare below cuff	No clothing beneath sphygmomanometer cuff
10	Back supported	Chair or elevated bedhead supports patient's back

BP indicates blood pressure.

*StrideBP.org or validatebp.org^{12,13}

†Calibration in this context refers to verification of pressure transducer accuracy against a reference standard and functional testing, as applicable to the device type. For oscillometric devices, this includes biomedical engineering verification of pressure measurement accuracy; for aneroid devices, this includes traditional calibration against a standardized reference at specified intervals.



descriptive findings of the audit is outlined in Tables 2 and 3.

Of the 278 observed measurements, 1 patient had bilateral upper limb lymphoedema precluding upper-arm measurement; BP was measured on the forearm with an incorrectly sized cuff. Eight patients were dependent for all bed mobility: 1 had an acute hip fracture making repositioning inappropriate, while the remaining 7 would have required 2-person assistance for optimal positioning, which was not performed. One patient with dementia was agitated and uncooperative with arm positioning during measurement. All measurements were included in the analysis as not meeting consensus criteria where applicable.

The median age of observed patients was 70 years (interquartile range, 61–84), and 56% were female. The comorbidity burden was high, with a median Charlson Comorbidity Score of 6 (interquartile range, 4). Forty-three percent had a documented history of hypertension, and 17% had documentation of atrial fibrillation, atrial flutter, or a conduction defect requiring internal pacing.

As the audited BP measurements were part of routine vital sign observation rounds, heart rate, temperature, pulse oximetry, respiratory rate, and pain scores were also assessed simultaneously in 275 patients. It took each observer an average of 2 minutes and 11 seconds to collect all vital signs. Nurses, their students, and their assistants were included in the audit. Fifty-one percent of the observers were registered nurses or clinical nurse consultants (attained via a 3-year university degree-based qualification), 36% were student nurses and the

remainder were assistants in nursing (12%) or enrolled nurses (1%), the latter 2 qualifications reflecting 12 to 24 months at a vocational college, respectively (Table 2). The mean first BP was 126/72±20/10 mm Hg (mean ± SD), with a mean heart rate of 82 bpm (SD, 15). Twenty-three BP measurements were outside the locally defined normal clinical parameters.²⁰ Five of these 23 abnormal readings were not remeasured before documentation in the electronic medical record. Of the 18 BP measurements that were outside the normal clinical parameters, 11 (61%) corrected to the normal reference parameters with no intervention other than repeating the BP measurement (Table S2).

The type of cuff used varied based on nursing unit manager procurement as it is the responsibility of each ward to order their own cuffs, as opposed to the BP devices themselves whose purchasing and maintenance were the responsibility of the hospital's biomedical engineering department. The surgical ward allocated a disposable Welch Allyn (WA) cuff to each patient on admission. This was then used for each patient for the length of stay and kept attached to the end of the patient's bed. The other 2 medical wards had 75% disposable and 25% reusable cuffs, all WA. Disposable cuffs accounted for 82% of cuffs for the 278 BP measurements.

Four different cuff sizes were observed in use, with 78% performed using the adult size cuff (for arm circumference range 25–34 cm), small adult cuffs in 17% (range 20–26 cm), large adult cuffs in 5% (range 32–43 cm), and a child cuff in <1% of BP measurements (range

Table 2. Environmental, Procedural, and Device Characteristics

Environmental characteristics	n	(%)
Observed blood pressure measurements	278	100
Number of observed patients by ward	153	
Surgical ward	85	31
Acute Geriatrics ward	92	33
Hematology/Oncology ward	101	36
Time of day observed		
Morning	127	46
Midday/early afternoon	151	54
Room type		
Single	73	26
Two-bed	45	16
Four-bed	160	58
Procedural characteristics		
Observer qualification		
Student, n (%)	100	36
Assistant in nursing, n (%)	32	12
Enrolled nurse, n (%)	3	1
Registered nurse, n (%)	140	50
Clinical nurse consultant, n (%)	3	1
Simultaneous measurement of other vitals	275	99
Medications given in the last 30 min	17	6
Time taken for collection of observation, min; mean (SD)	2.1	1.5 (SD)
Patient position		
Seated	72	26
Standing	1	0.4
Supine	78	28
Right lateral	8	3
Left lateral	4	1
Seated in bed/Fowler's position	115	41
Left arm selected limb	139	50
Clothing on limb		
Bare arm	180	65
Thin sleeve	69	25
Thick Sleeve	22	8
Sleeve bunched up above cuff	7	3
Cuff type		
Reusable	49	18
Disposable	229	82
Talking during procedure		
Observer talking	204	73
Patient talking	148	53
Any talking	214	77
Correct cuff position	189	68
Cuff appropriate tightness	258	93
Cuff appropriate position AND tightness	181	65
Number of BP readings per encounter		

(Continued)

Table 2. Continued

Environmental characteristics	n	(%)
1	259	93
2	11	4
3	8	3
Feet supported		
Yes	67	24
No	6	2
NA	205	74
Device characteristics		
Model for primary measurement		
Welch Allyn 75CE CONNEX	273	98
Calibration within the last 12 mo	256	92

BP indicates blood pressure.

15–21 cm). While no staff were observed measuring patient arm circumference, 77% of BP measurements were confirmed to have been performed using the correct cuff size (Tables 2 and 3).

Only 10 readings out of 278 were conducted using manual auscultation, and all used aneroid devices. None of these devices had a record of having undergone appropriate calibration. On general inspection of all aneroid devices by the independent auditor, there were no observable issues with the integrity of the tubing and casing of the BP devices. One of the 10 auscultatory BP measurements was a predetermined decision to conduct auscultatory readings on a particular patient due to concerns about the accuracy of previous automated readings. The remaining 9 were conducted as a second measurement after an aberrant reading using an automated device. Seven of the 10 auscultatory BP measurements had correct stethoscope position. All auscultatory BP measurements were done with a preceding palpated radial artery estimate of the SBP before proceeding to the auscultatory measurement.

BP Measurement Device Validation

Five device models were observed during the audit: 4 automated oscillometric devices and 1 aneroid device (WA 767, used for a single measurement with no available calibration data). The majority of measurements (273/278, 98%) were conducted using the WA 75CE CONNEX vital signs monitor, with the remainder distributed across 3 other devices: WA Spot LXI (n=2), WA Spot 42N0B (n=1), and CRITICARE 506DN (n=1).

Only the WA 75CE CONNEX and Spot LXI have published validation data, confirmed through the Stride BP database when used with reusable cuffs.^{12,13} However, no validation data exist for these devices with disposable cuffs. Of the 275 measurements performed on these 2 validated devices, only 49 (18%) used reusable cuffs, while 226 (82%) used disposable cuffs. Validation could

Table 3. Patient Characteristics

Patient characteristics	n	(%)
Female	157	57
Age, y; median (IQR)	73	23 (IQR)
19–49	40	15
50–69	78	28
70–89	132	48
>90	28	10
Established diagnosis of hypertension	119	43
Arrhythmia	47	17
Charlson comorbidity index, median (IQR)	6	4 (IQR)
Arm circumference, cm; mean (SD)	28	4.9 (SD)
Mean 1st SBP, mmHg; SD	126	20 (SD)
Mean 2nd SBP, mmHg; SD	105	26 (SD)
Mean 3rd SBP, mmHg; SD	105	26 (SD)
Mean heart rate, bpm; SD	82	15(SD)

IQR indicates interquartile range; mmHg, millimeter of mercury; and SBP, systolic blood pressure.

not be confirmed for the remaining 3 devices through published studies from manufacturers or independent validation organizations.^{12,13}

Consequently, when considering the validated device and cuff combination as the validated unit, only 49/278 (18%) of all observed BP measurements were performed using fully validated equipment.

Concordance of BP Measurements With Consensus Criteria

The proportion of observed BP measurements achieving the desired 10 consensus criteria is reported in Figure 2, in order of ranked importance as per expert consensus. None of the measurements met all 10 criteria. Of all BP measurements with an automated device, 92% were performed with a recently calibrated device, uncrossed legs in 81%, 77% had the correct cuff size, and 75% had a supported back. Twenty-three percent met the criterion of no talking during the BP measurement. Seven percent of measurements included ≥ 2 readings, all of which were prompted by an initial reading falling outside the hospital's designated normal parameters. Not unexpectedly for an inpatient population, most patients were either supine (28%) or seated in bed (41%), rather than seated in a chair (26%). One patient was standing (0.4%), 8 were in right lateral (2.9%), and 4 in left lateral recumbency (1.5%). Sixty-four percent had a bare arm below the BP cuff, and a further 25% had only a thin sleeve (Table 2).

Subgroup Analysis for Independent Predictors of Meeting Consensus Criteria

Subgroup analyses assessed whether arm circumference, ward, systolic BP category, and observer

experience were associated with meeting each of the 10 consensus criteria (Figure 3). Mid-range arm circumference (26–29 cm) was associated with higher use of the correct cuff size ($P<0.001$), whereas smaller arms (17–25 cm) correlated with higher correct cuff positioning ($P=0.003$). Across wards, the surgical ward location had 100% compliance with a calibrated device (estimate=1.00) and performed better on several procedural aspects compared with the medical wards, namely device calibration ($P<0.001$), cuff at heart level ($P<0.001$), and back supported ($P=0.03$). The Aged Care ward was less likely to have a bare arm below the cuff ($P=0.03$). An SBP <100 mmHg was associated with taking ≥ 2 measurements ($P<0.001$). By observer experience, senior staff had lower compliance with no talking during measurement (either the patient or observer; $P=0.006$). Students were less likely to have a correct cuff position ($P=0.014$) or to have the back supported ($P=0.035$), but more likely to use calibrated devices ($P<0.001$).

DISCUSSION

In this audit of 278 BP measurements, no measurement met all 10 expert-consensus criteria for accurate technique. Critically, only 7% of measurements included more than 2 readings—the criterion ranked most important by our expert panel. In addition, only 18% of measurements used a validated device with a validated cuff.

Our findings build on a prior 2014 United Kingdom audit of 100 BP measurements in medical inpatients using outpatient BP methodology standards.²¹ Both studies showed poor compliance for obtaining multiple readings (4% in the UK study versus 7% in the present study) and appropriate cuff sizing (64% versus 77%, respectively). The UK study reported less talking during measurement (41%) compared with the present study (77%), whereas both audits demonstrated high adherence to proper patient positioning. These consistent patterns across settings suggest our observations reflect more than site-specific challenges.

The low rate of validated device and cuff use (18%) was driven by the majority use of unvalidated disposable cuffs. The WA disposable cuffs, designed for single-patient use with guaranteed accuracy for 250 inflations, were reused across patients in 2 of 3 wards without tracking of inflation cycles. This represents both a significant source of measurement uncertainty and an actionable target for quality improvement.

Talking was more frequent during senior staff assessments, possibly reflecting multitasking behaviors when combining medication and observation rounds, or patients perceiving senior staff as more likely to have information about their treatment. SBP outside normal clinical ranges was the only prompt for >1 measurement. Patients with outlier arm circumference were less likely to receive



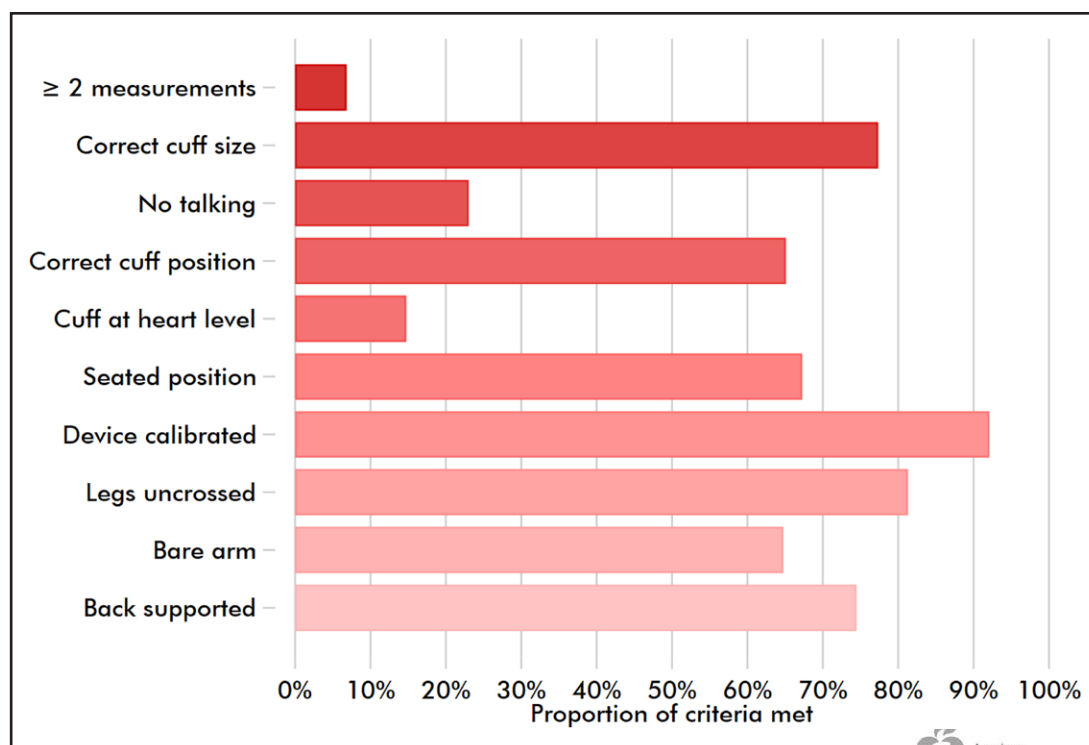


Figure 2. Concordance of observed blood pressure (BP) measurement technique with consensus criteria for accurate inpatient BP measurement.

The figure details the proportion of each of the 278 BP measurements that met each of the consensus criteria for accurate BP measurement. They are listed in order of ranked importance as per the Inpatient BP expert panel, with the most important criteria at the **top**, and the least important at the **bottom**, as indicated by the color gradient.

correct cuff sizing, potentially reflecting limited availability of nonstandard cuff sizing. Cuff positioning was better for those with smaller arm circumference, which may have reflected the correlation between smaller arm circumference and a more cylindrical arm shape.

Patients with acute illness need tailored BP protocols relevant to inpatient care. Acute inpatients often require supine or seated-in-bed positioning, and emerging evidence suggests some office-based requirements (ambient noise restrictions²² and thin sleeve removal²³) may not significantly impact accuracy. All BP measurements were included regardless of the clinical feasibility of consensus-concordant technique. While only 1 patient had absolute anatomic contraindications to upper-arm BP measurement, practical barriers existed: 8 patients had mobility limitations, and 1 was behaviorally uncooperative. Thus, only 1% of observed measurements had absolute contraindications to adherence to consensus criteria. In the acute inpatient setting, postponing or omitting vital sign measurements due to suboptimal conditions is clinically inappropriate, as timely BP measurement is integral to early detection of patient deterioration. A further 41% of measurements were taken seated in bed, which, if considered a valid position for measurement for inpatients would increase concordance with the criterion from 26% to 67%. The WHO now permits thin sleeves under the cuff in its technical specification for

accurate office BP measurement,⁴ supporting pragmatic protocol adaptation. We also noted that despite a 17% prevalence of documented clinically significant arrhythmia, no consideration of this comorbidity was given to the choice of BP device, despite this arrhythmia's known impact on the reliability of oscillometric readings.²⁴

Standardized, high-quality BP measurement in hospitals carries significant clinical importance. Elevated inpatient BP strongly predicts antihypertensive intensification at discharge, affecting 1 in 7 older patients admitted for noncardiac conditions⁶ yet this intensification increases short-term risks without long-term cardiovascular benefit.⁵ Those studies highlighted a discrepancy between office-based and inpatient BP assessments.^{5,25} One explanation is that inpatient BP measurement quality resembles that observed in our audit, risking inaccurate data. Given that 20% to 30% of adult hospital admissions (>2 million annually in Australia) involve comorbid hypertension, ensuring measurement accuracy has broad population implications.^{26,27}

This study has several strengths. It is the largest audit to characterize patient, procedural and device related characteristics that constitute a standard inpatient BP measurement, capturing a representative seniority mix of health care staff, and including both medical and surgical patients with a high comorbidity index of a standard tertiary hospital. There are also limitations. This was a

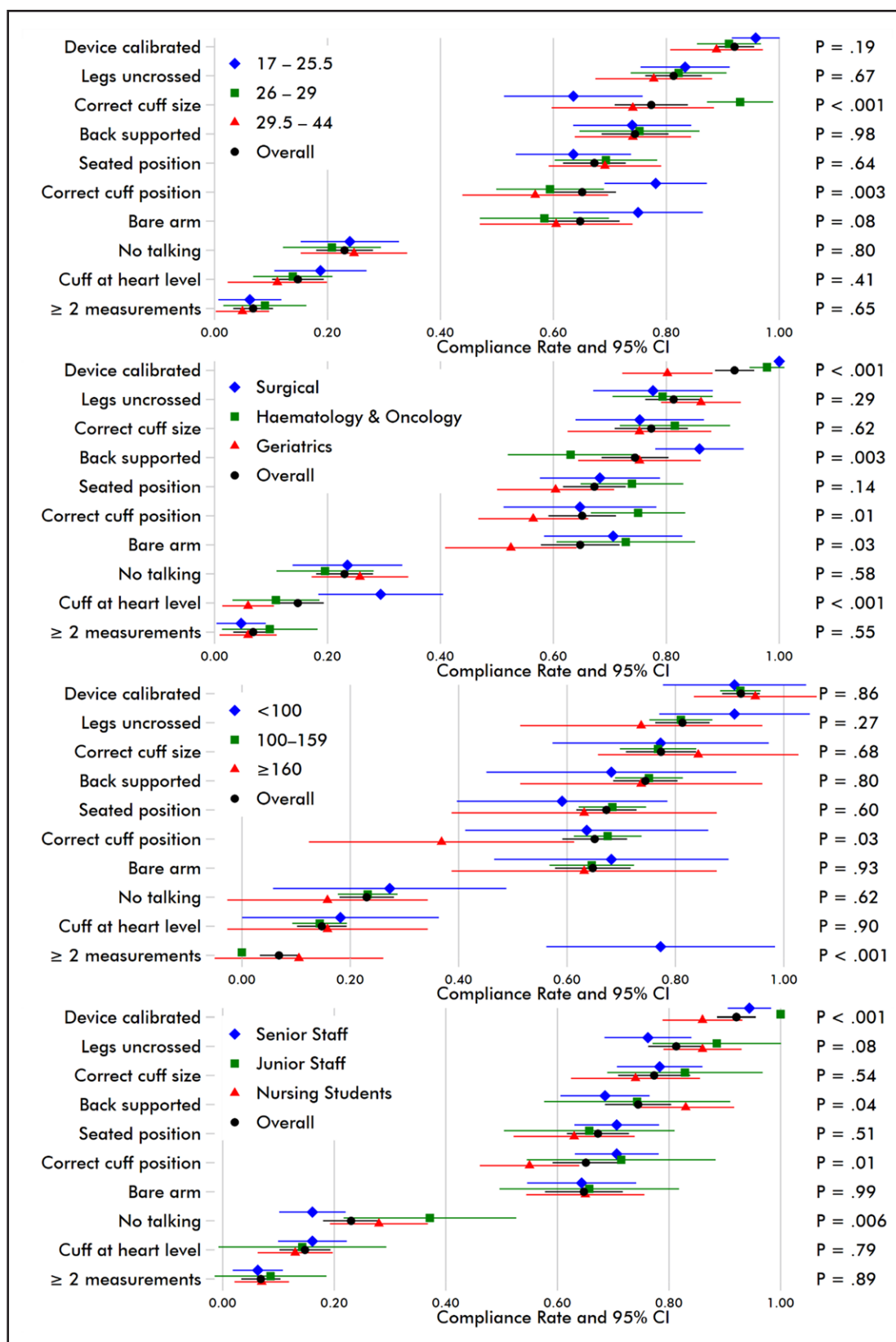


Figure 3. Concordance with consensus criteria according to (A) arm circumference, (B) ward, (C) systolic blood pressure (SBP), and (D) observer qualification.

The 10 consensus criteria are outlined here, in order of concordance. **A**, Arm circumference is grouped into 3 categories: 17 to 25.5 cm, 26 to 29 cm, and 29.5 to 44 cm. Groupings correlated to standard device cuff sizes with necessary variation to reflect there is overlap in cuffs. **B**, The 3 ward locations reflect usual patient population admitted to each location: acute surgical patients, (Continued)

Figure 3 Continued. hematological and oncological patients and acute geriatric patients to each of the respective 3 wards. **C,** SBPs are grouped into 3 to reflect a combination of the clinical response calling criteria for inpatients in the observed hospital and systolic criteria for grade 3 hypertension SBP <100 mmHg, SBP 100 to 159 mmHg and SBP >160 mmHg. **D,** Observer qualifications are categorized into 3 groups according to the level of training of each observer. Senior staff included registered nurses and clinical nurse consultants (university degree-based training), junior staff were enrolled nurses and assistants in nursing (vocational college education) and finally university nursing students.

single-center audit limited to 3 acute wards, and a larger multicenter audit might give a better understanding of the trends in standard practices in Australian tertiary hospitals. Night BP measurements have not been captured and likely vary considerably, particularly in reference to patient positioning. Our classification of adherence used subjectively determined thresholds (good >70%, moderate 30% to 70%, poor <30%) selected a priori based on audit methodology guidance.^{16–18} Different thresholds might paint a different picture of concordance. Techniques of doctors and allied health staff should also be characterized in future studies.

PERSPECTIVES

This audit reveals a critical gap between our expert consensus BP measurement criteria and routine hospital practice. The striking paucity of multiple readings—the criterion experts ranked as most important—alongside widespread use of unvalidated disposable cuffs suggests that current inpatient BP data may be systematically unreliable. This measurement uncertainty matters: elevated inpatient readings drive antihypertensive intensification, yet such intensification appears to increase short-term harm without long-term benefit. If hospital BP measurements are as technically flawed as our findings suggest, clinicians may be making consequential decisions based on artifactually altered values. The path forward requires 3 parallel efforts. First, the international community of BP management policymakers must develop evidence-based inpatient BP protocols that balance measurement rigor with clinical pragmatism. Second, validation studies of hospital-specific protocols and of disposable cuff systems are urgently needed, alongside standardization or transition to validated reusable equipment. Third, these protocols must be embedded into staff education and institutional quality frameworks. Only by ensuring measurement quality can we determine whether elevated inpatient BP represents a BP requiring intervention or a measurement artifact demanding restraint—a distinction with profound implications for patient safety and cardiovascular outcomes.

ARTICLE INFORMATION

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